# BreastGuard AI

*Explainable Breast Cancer Classification & Trustworthy AI Dashboard*

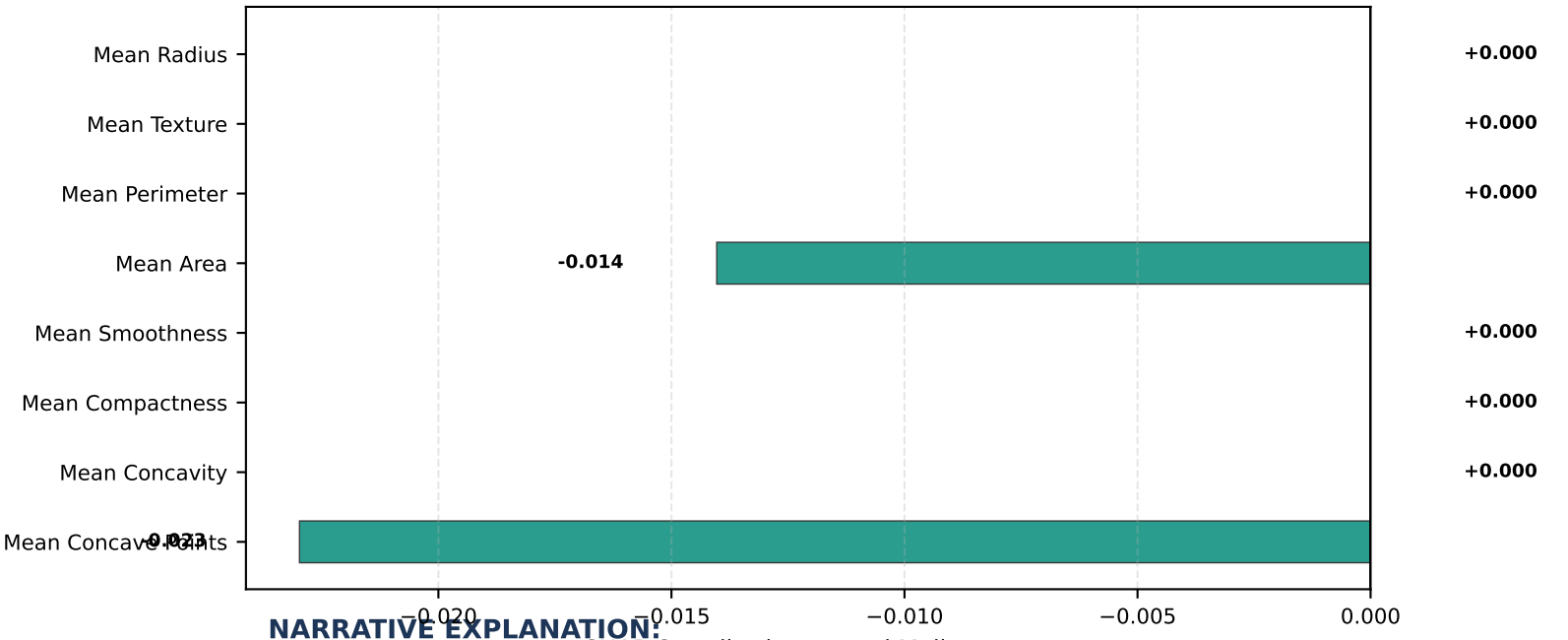
Report Date: 2026-11-06 14:08:07
Decision Threshold: 0.37

Model Architecture: Support Vector Machine
Risk Stratification: Low Risk (<20%)

**PREDICTED DIAGNOSIS:**

**BENIGN**

**P(Malignant): 3.0%**
**P(Benign): 97.0%**




## NARRATIVE EXPLANATION:

Model Decision: Classified as Benign with P(Malignant) = 3.0% (Operating Threshold = 0.37, Population Baseline

Features decreasing model output toward Benign:

- Worst Area (observed value: 686.500) contributed -0.047 toward the benign classification.

## MANDATORY CLINICAL & REGULATORY DISCLAIMER

*"This is an educational research prototype, not a clinically validated diagnostic system. It must not be used to diagnose cancer, make treatment decisions, or replace professional medical evaluation."*